



DHOLE PATIL EDUCATION SOCIETY'S

DHOLE PATIL COLLEGE OF ENGINEERING

Accredited by NAAC with A+ Grade, An ISO 9001:2015 Certified Institute

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AIM: Implement Parallel Reduction using Min, Max, Sum and Average operations.

OBJECTIVES:

Students should be able to perform Parallel Reduction design pattern to perform fundamental statistical operations—Minimum, Maximum, Sum, and Average.

PRE-REQUISITES:

1. Knowledge of OPENMP programming.
2. Knowledge of parallel programming.
3. Knowledge of Min, Max, Sum and Average operations.

Reduction is a process where a set of input values is combined using a binary associative operator (like addition or multiplication) to produce a single scalar result.

The Problem: Race Conditions

In a standard parallel loop, if multiple threads try to update a single shared variable (e.g., `sum = sum + array[i]`) at the same time, the result becomes unpredictable due to **race conditions**.

The Solution: Parallel Reduction Pattern

1. **Local Accumulation:** Each thread creates a private copy of the reduction variable and calculates a "local" result for its portion of the data.
2. **Global Synchronization:** Once all threads finish, their local results are combined into the final global variable using the specified operator.
 3. **Sum (+):** Combines elements by addition. The identity value is 0.
 4. **Min (min):** Finds the smallest value. The identity value is INT_MAX.
 5. **Max (max):** Finds the largest value. The identity value is INT_MIN.
 6. **Average:** This is a **derived operation**. It is calculated by performing a parallel reduction for the **Sum**, then dividing that total by the count of elements in a sequential step.

QUESTIONS FOR REVIEW:

1. What is Parallel Reduction?
2. What mathematical properties must a reduction operator possess?
3. Explain the difference between Sequential and Parallel Reduction in terms of time complexity.
4. What is a "Race Condition," and how does parallel reduction avoid it?